Smooth Sailing

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A 2023-2024 crash data analysis states

Drivers from ages 16-17 are 6x more likely to get into a car crash when compared to other age demographics.

The Problem

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With every year, statistics are only going to get worse as more and more people begin to drive, making it even more difficult for newer drivers

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As beginner drivers, they may not be able to completely process all the information required to drive a car safely, leading them to not keep their speed in check.

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When accelerating, braking, or turning their car too harshly, it can end up damaging or wearing out certain components, but more importantly, it endangers other drivers on the road along with any passengers in the car.

Our Intentions

Plan

Our team plans to design and create a prototype that will help to mitigate this issue that can later be turned into an easily accessible product that can help anyone who needs it

Prototype

Using an Arduino connected to an accelerometer, LEDs, and a piezo buzzer, we created a prototype that can accurately measure the acceleration and G-forces of the car and alert the driver when it accelerates or brakes too quickly

Video

We intend to show proof of concept with a video of a working prototype within a car itself, showing that our product can truly aid in improving new driver's skills.

Target Consumers

1st

Our target consumers would be either 15+ year-old kids who just began learning to drive or anyone else who feels that they need more practice driving and want a product to help them improve.

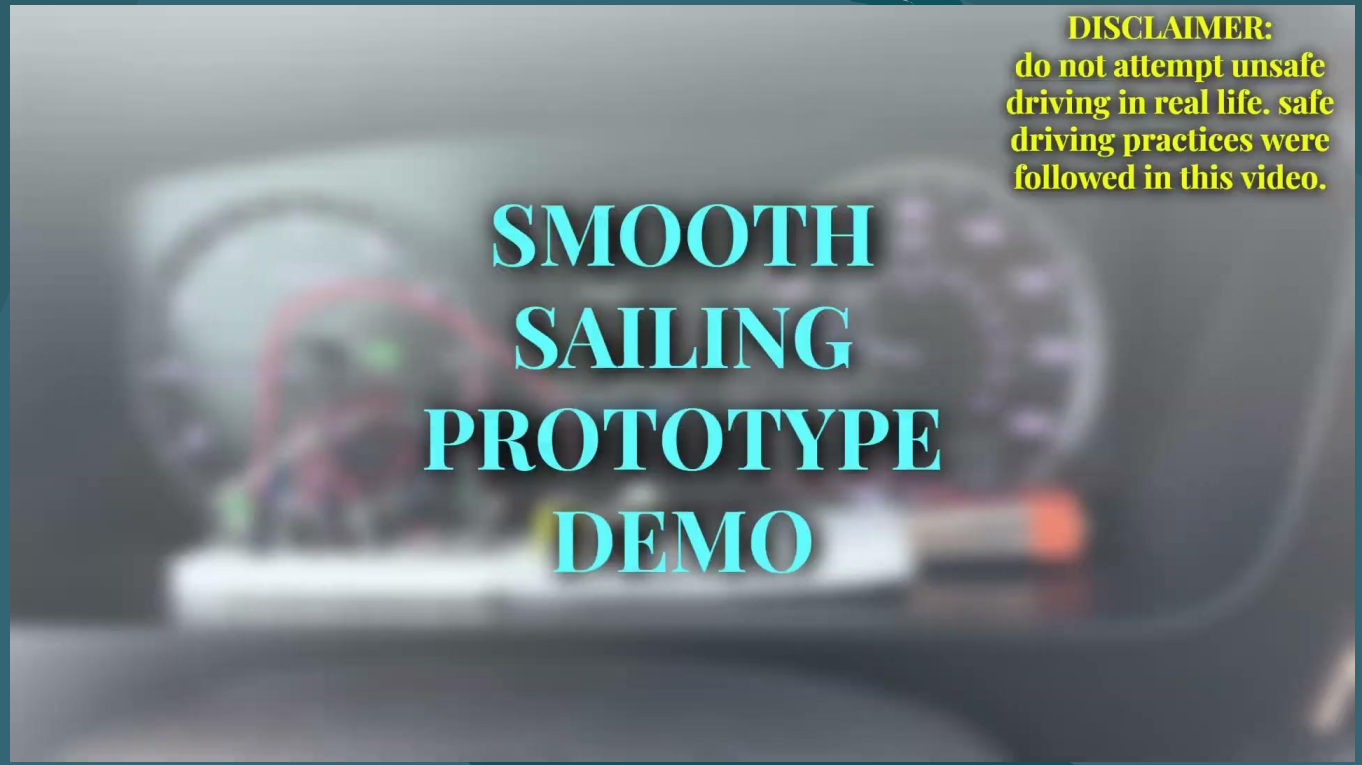
2nd

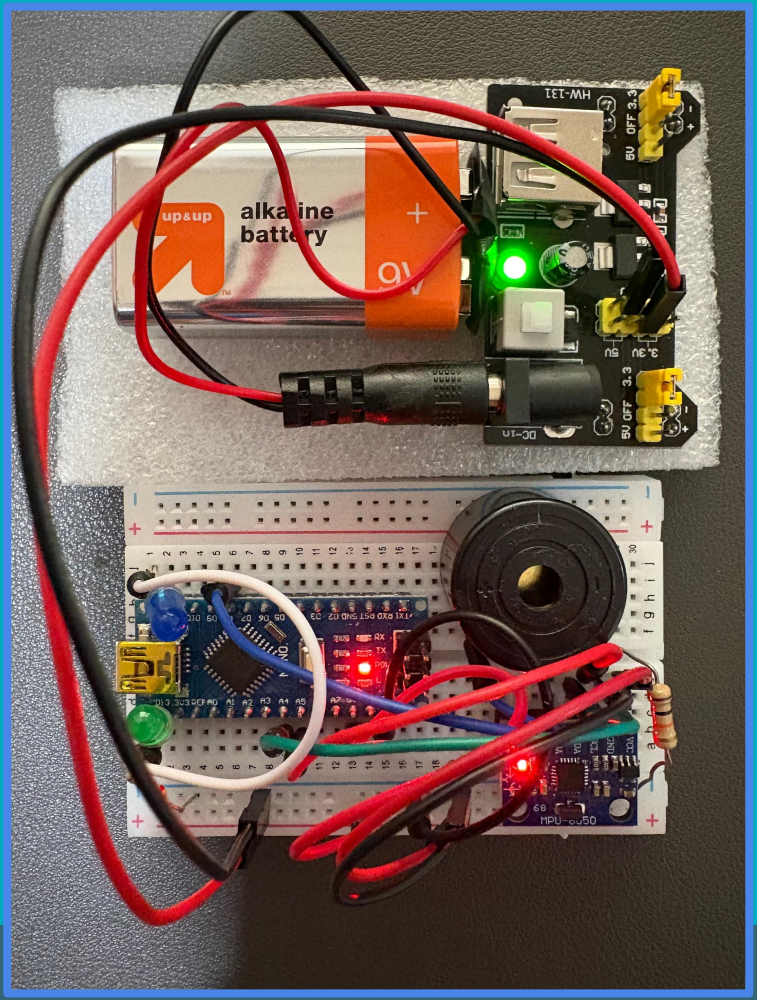
New drivers especially may not be able to properly tell their speed since they're not entirely used to driving yet, so they would greatly benefit from a device that lets them know if they begin to accelerate or brake too suddenly.

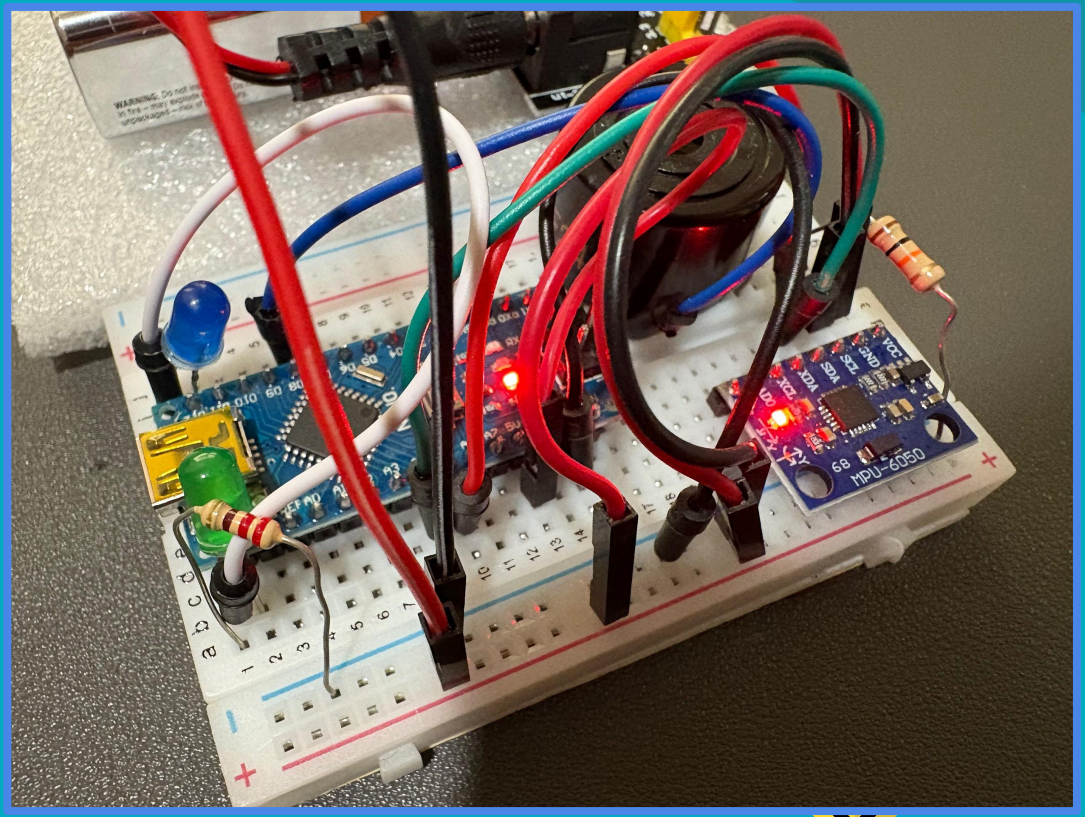
3rd

The parents of beginner drivers may also buy the product if they believe that their child needs more help with driving practice.

Our Solution




up&up
alkaline
battery
9V
HW-131
MPU-6050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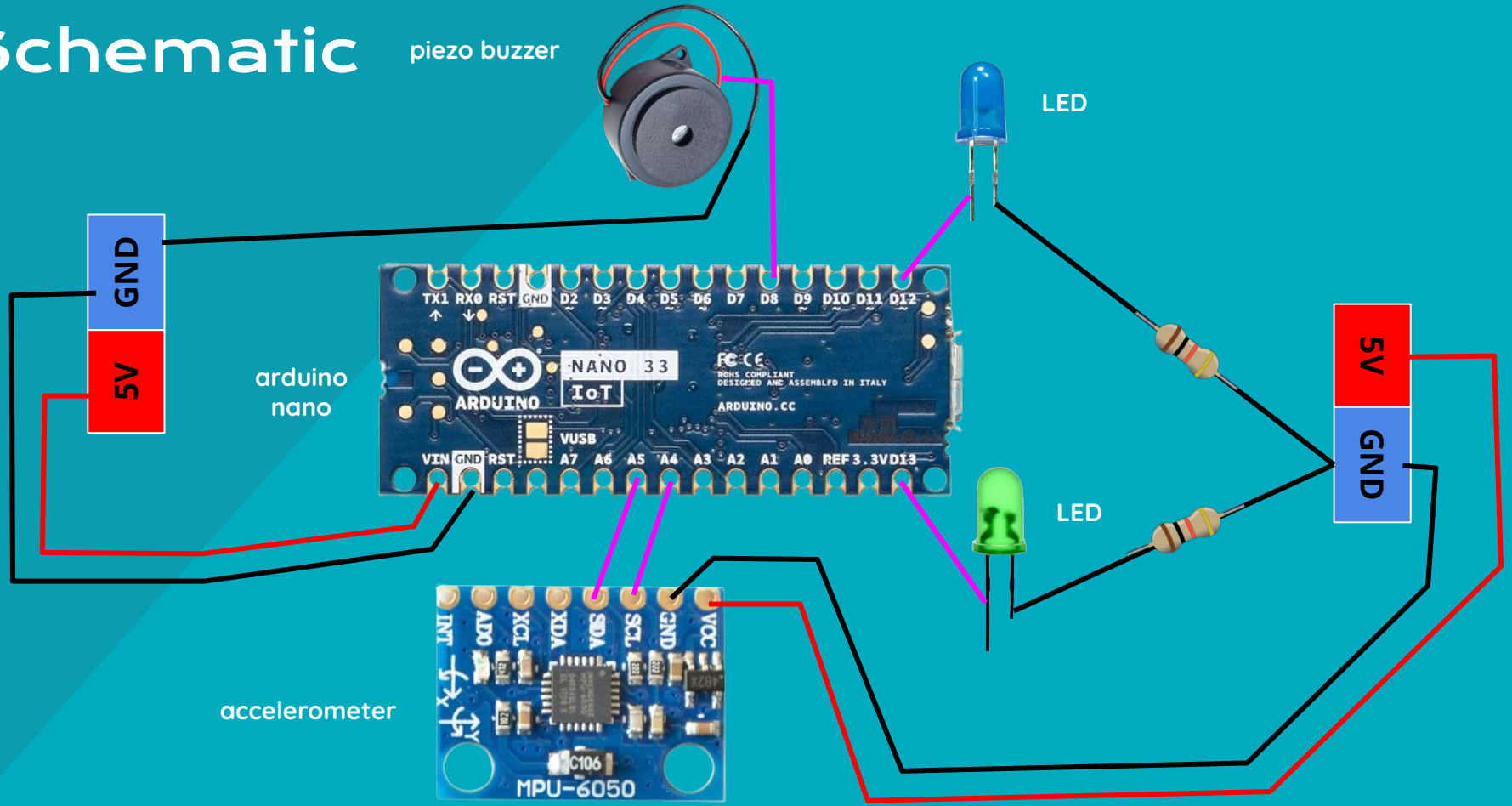
Schematic


piezo buzzer
LED
GND
5V
arduino
nano
TX1 RX0 RST GND D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12
NANO 33
IoT
ARDUINO
VUSB
VIN GND RST A7 A6 A5 A4 A3 A2 A1 A0 REF 3.3V D13
ARDUINO.CC
5V
GND
LED
accelerometer
INT AD0 XCL XDA SDA SCL GND VCC
C106
MPU-6050

Solution Evaluation

Expenses

total cost: $14.24 per unit to produce

price breakdown:
$5.33 per Arduino Nano
$1.33 per small breadboard
$0.33 per unit for wires/leds/resistors
$5.50 per accelerometer
$1.75 per battery pack

We will sell the SmoothSailing module for $30, with a 53% profit margin

Pros

- Beeps are easy to differentiate
- Don't have to divert attention from the road to notice error
- Low cost so it's easily accessible

Cons

- No set place to mount the device in the car
- Takes up a decent amount of space
- No way to look back on data and track improvement

Thank You